Submitted By:

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CSE Cyber Security

3rd/6th

Assignment – 3 (Data Science)

Q1. Write a Python program to draw line charts of the financial data of

Alphabet Inc. between October 3, 2016 to October 7, 2016.

Sample Financial data (fdata.csv):

Date,Open,High,Low,Close

10-03-16,774.25,776.065002,769.5,772.559998

10-04-16,776.030029,778.710022,772.890015,776.429993

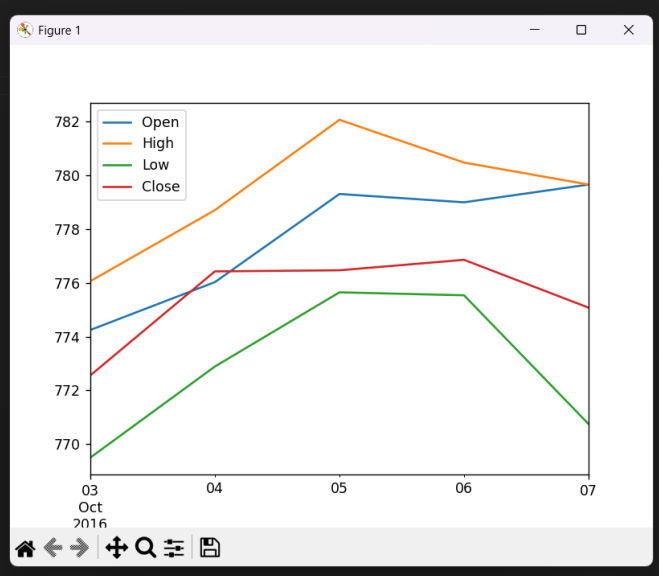
10-05-16,779.309998,782.070007,775.650024,776.469971

10-06-16,779,780.47998,775.539978,776.859985

10-07-16,779.659973,779.659973,770.75,775.080017

Ans.

```
import matplotlib.pyplot as plt
import pandas as pd
df = pd.read_csv('fdata.csv', sep=',', parse_dates=True, index_col=0)
df.plot()
plt.show()
```



Q2. Write a Python program to Program to show number of students vs

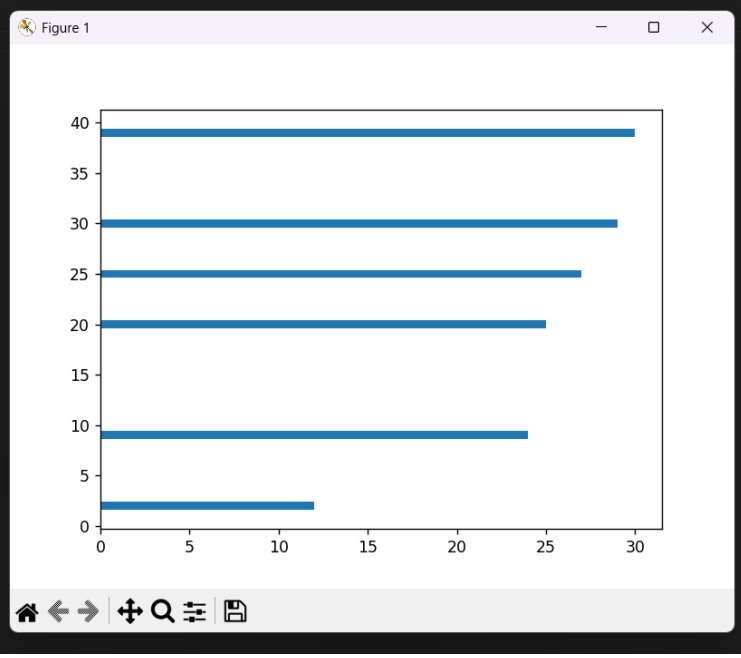
marks obtained (Horizontal Bar Plot)

nos = [2, 9, 20, 25, 30, 39]

marks = [12, 24, 25, 27,29, 30]

Ans.

```
import matplotlib.pyplot as plt
nos = [2, 9, 20, 25, 30, 39]
marks = [12, 24, 25, 27, 29, 30]
plt.barh(nos, marks)
plt.show()
```



Q3.

#Example 6: The file "monthsales.csv" have stored the sales (in Rs) made in first six months for four years.

2018	2019	2020	2021
2500	2600	2450	3250
2200	1800	2000	3200
3000	2500	3000	3800
2800	3000	3100	2800
2750	3200	1950	2900
3500	2000	3550	2300

Year-wise sales of first six months

Draw the line chart for the data given above with following details.

1. **Chart title as "Year-Wise Sales"**
2. **X-axis label as "Months"**
3. **Y-axis label as "Sales"**

Ans.

```python
import pandas as pd
import matplotlib.pyplot as plt
# reads "monthsales.csv" to df by giving path to the file
df=pd.read_csv("monthsales.csv")
#create a line plot of different color for each week
df.plot(kind='line', color=['red', 'blue', 'brown', 'Yellow'])
# Set title to "Year-Wise Sales"
plt.title('Year-Wise Sales')
# Label x axis as "Months"
plt.xlabel('Months')
# Label y axis as "Sales"
plt.ylabel('Sales')
#Display the figure
plt.show()
```

Q4. Assuming the same CSV file, i.e., monthsales.csv, plot the line chart with following customisations. Chart should have Month name on X-axis instead of numbers.

Maker ="*"

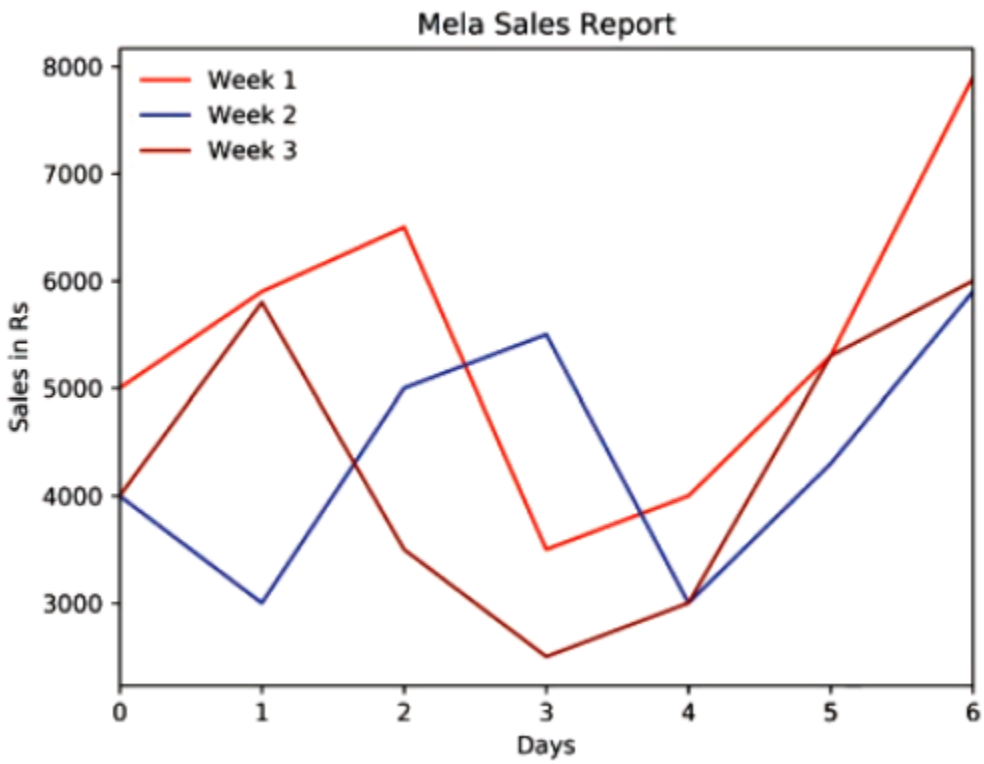
Marker size=10

linestyle="–"

Linewidth =3

Ans.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the data from the CSV file
df = pd.read_csv('monthsales.csv')

# Plotting the line chart with specified customizations
# Marker='*', Marker size=10, Linestyle='--', Linewidth=3
plt.plot(df['Month'], df['Sales'], marker='*', markersize=10, linestyle='--', linewidth=3) df =    2018  2019  2020  2021 0  2500  2600  2450  3250 1

# Adding labels and title for clarity
plt.xlabel('Month')
plt.ylabel('Sales')
plt.title('Monthly Sales Report')

# Display the plot
plt.show()
```



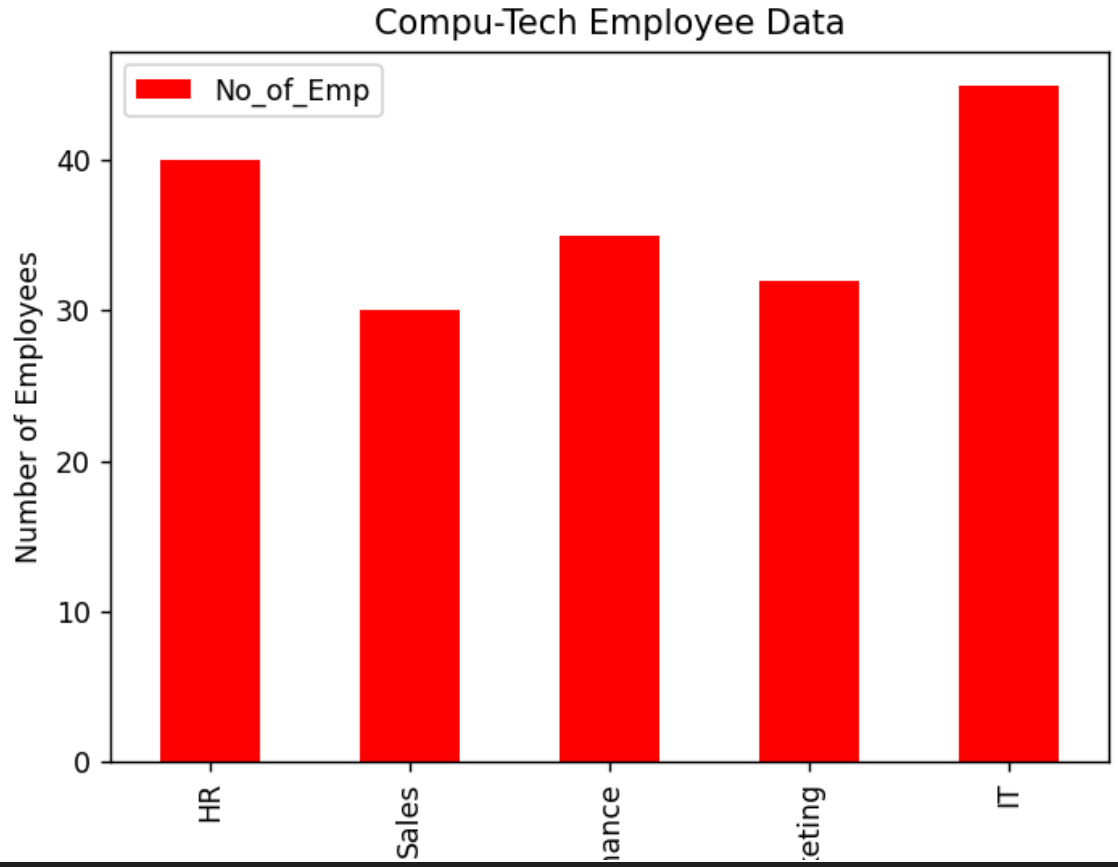
Q5.

Anant is working as a Data Analyst in a company named “”Compu-Tech”. He stored the following data in a file named “emp.csv”. He wants to present “Department” (First Column) and “No_of_Emp” (Third Column) in a bar chart. Help him to write the code.

Department	Expenditure (in Rs)	No_of_Emp
HR	2500	40
Sales	3000	30
Finance	2200	35
Marketing	5100	32
IT	3900	45

- **Chart Title: Compu-Tech Employee Data**
- **X-Axis Label: Departments**
- **Y-Axis Label: Number of Employees**
- **Department should be displayed on X-axis**
- **Colour of bar should be RED**

Ans.




```
import matplotlib.pyplot as plt
import pandas as pd
df = pd.read_csv("monthsales.csv", usecols = ['Department', 'No_of_Emp'])
df.plot(kind = "bar", x = "Department", color = "red", )
plt.title("Compu-Tech Employee Data")
plt.xlabel("Departments")
plt.ylabel("Number of Employees")
plt.show()
```

Q6. Visit data.gov.in , search for the following in “catalogs” option of the website:

• Final population Totals, India and states

• State Wise literacy rate Download them and create a CSV file containing population data and literacy rate of the respective state.

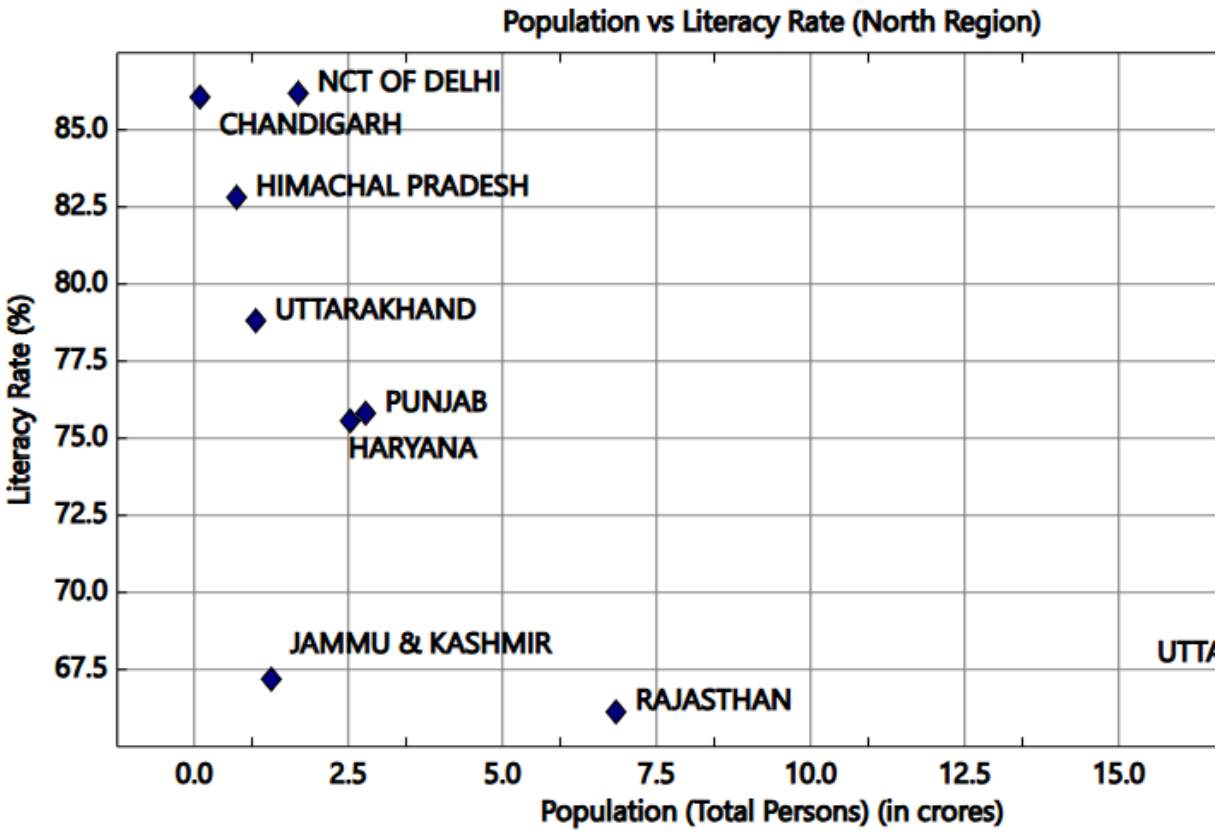
Also add a column Region to the CSV file that should contain the values East, West, North and South.

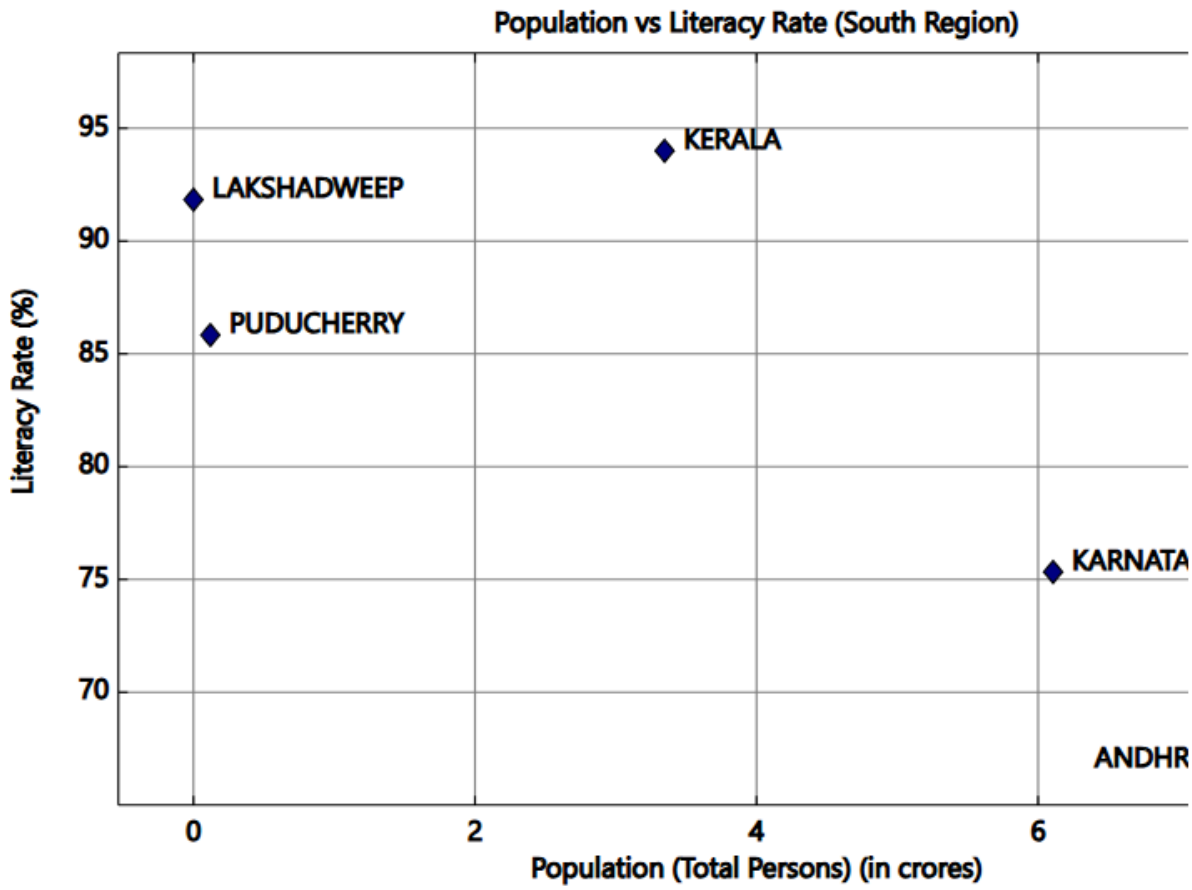
Plot a scatter plot for each region where X axis should be population and Y axis should be Literacy rate. Change the marker to a diamond and size as the square root of the literacy rate. Group the data on the column region and display a bar chart depicting average literacy rate for each region.

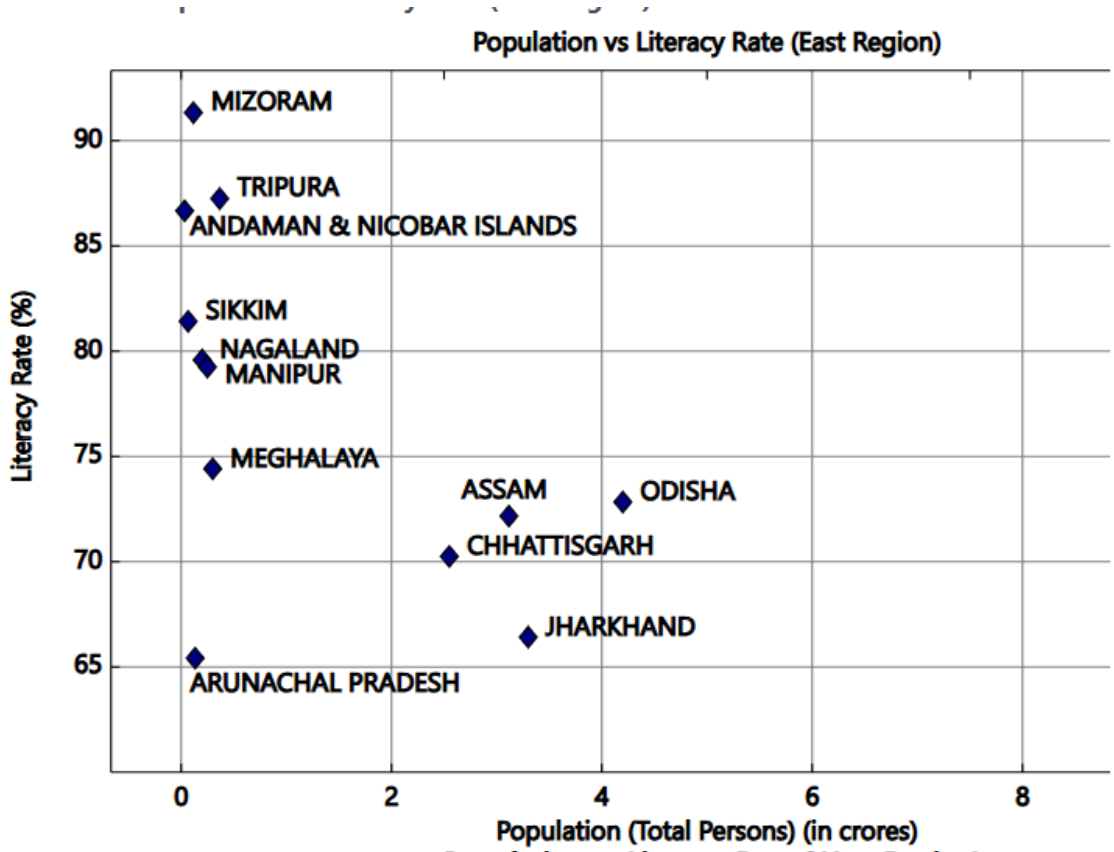
Ans.

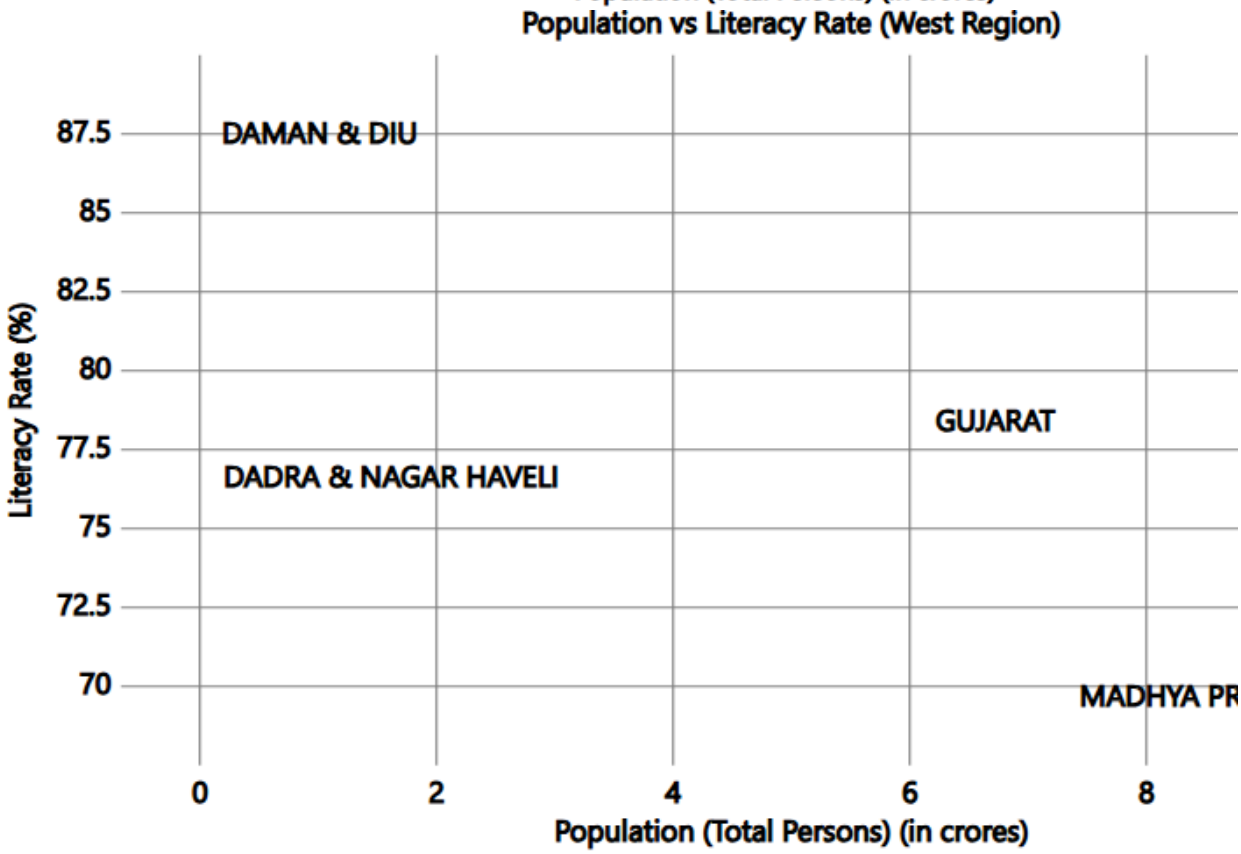
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# ----------------------------
# 1) Read the Census PCA file
# ----------------------------
# Put your file path here
path = "PCA0000_2011_MDDS.xls"

# The sheet name in this file is typically "PCA"
df = pd.read_excel(path, sheet_name="PCA")

# ------------------------------------------------
# 2) Keep only STATE level totals (TRU = Total)
# ------------------------------------------------
state_df = df[(df["Level"] == "STATE") & (df["TRU"] == "Total")].copy()

# ------------------------------------------------------------
# 3) Build population + compute literacy rate (Census logic)
#    Literacy Rate (%) = Literates (7+) / Population (7+) * 100
# ------------------------------------------------------------
state_df["Population"] = state_df["Total Population Person"]
state_df["Pop_0_6"] = state_df["Population in the age group 0-6 Person"]
state_df["Pop_7_plus"] = state_df["Population"] - state_df["Pop_0_6"]

state_df["Literates_7_plus"] = state_df["Literates Population Person"]
state_df["LiteracyRate"] = (state_df["Literates_7_plus"] / state_df["Pop_7_plus"]) * 100

# -------------------------------------
# 4) Add Region column (E/W/N/S mapping)
# -------------------------------------
region_map = {
    # North
    "JAMMU & KASHMIR": "North",
    "HIMACHAL PRADESH": "North",
    "PUNJAB": "North",
    "CHANDIGARH": "North",
    "UTTARAKHAND": "North",
    "HARYANA": "North",
    "NCT OF DELHI": "North",
    "RAJASTHAN": "North",
    "UTTAR PRADESH": "North",

    # East (includes North-East + eastern India)
    "BIHAR": "East",
    "SIKKIM": "East",
    "ARUNACHAL PRADESH": "East",
    "NAGALAND": "East",
    "MANIPUR": "East",
    "MIZORAM": "East",
    "TRIPURA": "East",
    "MEGHALAYA": "East",
    "ASSAM": "East",
    "WEST BENGAL": "East",
    "JHARKHAND": "East",
    "ODISHA": "East",
    "CHHATTISGARH": "East",
    "ANDAMAN & NICOBAR ISLANDS": "East",

    # West
    "MADHYA PRADESH": "West",
    "GUJARAT": "West",
    "DAMAN & DIU": "West",
    "DADRA & NAGAR HAVELI": "West",
    "MAHARASHTRA": "West",
    "GOA": "West",
```

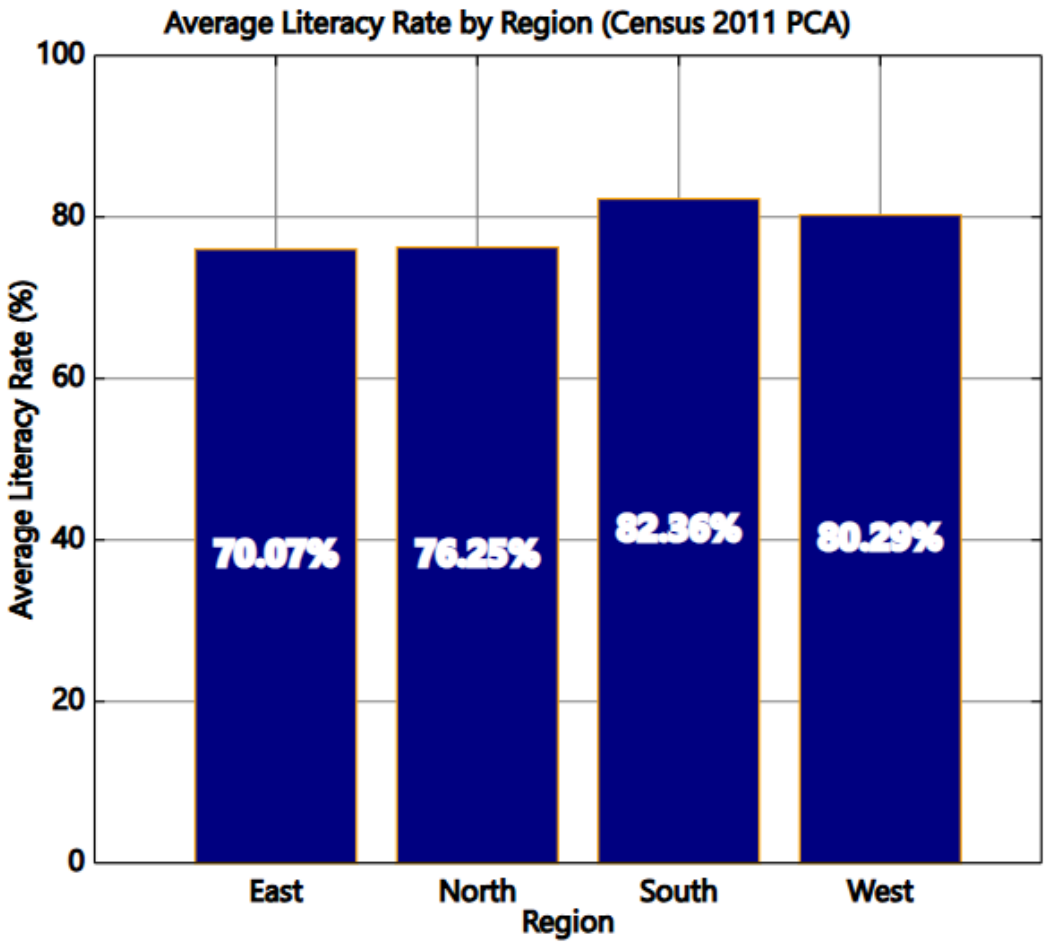
```python
    "LAKSHADWEEP": "South",
    "KERALA": "South",
    "TAMIL NADU": "South",
    "PUDUCHERRY": "South",
}

state_df["Region"] = state_df["Name"].map(region_map).fillna("North")

# ------------------------------------
# 5) Final table + export CSV
# ------------------------------------
final_df = state_df[["Name", "Population", "LiteracyRate", "Region"]].copy()
final_df.rename(columns={"Name": "State"}, inplace=True)

final_df["Population"] = final_df["Population"].astype("int64")
final_df["LiteracyRate"] = final_df["LiteracyRate"].round(2)

final_df.to_csv("state_population_literacy_region.csv", index=False)
print("Saved: state_population_literacy_region.csv")

# ------------------------------------
# 6) Scatter plots (one per region)
# ------------------------------------
regions = ["North", "South", "East", "West"]

for r in regions:
    sub = final_df[final_df["Region"] == r].copy()

    plt.figure(figsize=(8, 5))
    plt.scatter(
        sub["Population"],
        sub["LiteracyRate"],
        marker="D",                      # diamond marker
        s=np.sqrt(sub["LiteracyRate"])  # size = sqrt(literacy rate)
    )

    plt.title(f"Population vs Literacy Rate ({r} Region)")
    plt.xlabel("Population (Total Persons) (in crores)")
    plt.ylabel("Average Literacy Rate (%)")

    # Optional: label each point with the state name
    for _, row in sub.iterrows():
        plt.annotate(
            row["State"],
            (row["Population"], row["LiteracyRate"]),
            fontsize=7,
            xytext=(3, 3),
            textcoords="offset points"
        )

    plt.tight_layout()
    plt.show()

# ------------------------------------
# 7) Bar chart: average literacy by region
# ------------------------------------
avg_lit = final_df.groupby("Region", as_index=False)["LiteracyRate"].mean()

plt.figure(figsize=(7, 4.5))
plt.bar(avg_lit["Region"], avg_lit["LiteracyRate"])

plt.title("Average Literacy Rate by Region (Census 2011 PCA)")
plt.xlabel("Region")
plt.ylabel("Average Literacy Rate (%)")
```


Population vs Literacy Rate (North Region)
NCT OF DELHI
CHANDIGARH
HIMACHAL PRADESH
UTTARAKHAND
PUNJAB
HARYANA
JAMMU & KASHMIR
RAJASTHAN
UTTA
85.0
82.5
80.0
77.5
75.0
72.5
70.0
67.5
Literacy Rate (%)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
Population (Total Persons) (in crores)


Population vs Literacy Rate (South Region)
Literacy Rate (%)
95
90
85
80
75
70
KERALA
LAKSHADWEEP
PUDUCHERRY
KARNATA
ANDHR
0
2
4
6
Population (Total Persons) (in crores)


Population vs Literacy Rate (East Region)
MIZORAM
TRIPURA
ANDAMAN & NICOBAR ISLANDS
SIKKIM
NAGALAND
MANIPUR
MEGHALAYA
ASSAM
ODISHA
CHHATTISGARH
JHARKHAND
ARUNACHAL PRADESH
Literacy Rate (%)
90
85
80
75
70
65
0
2
4
6
8
Population (Total Persons) (in crores)


Population vs Literacy Rate (West Region)
DAMAN & DIU
GUJARAT
DADRA & NAGAR HAVELI
MADHYA PR
Literacy Rate (%)
87.5
85
82.5
80
77.5
75
72.5
70
0
2
4
6
8
Population (Total Persons) (in crores)



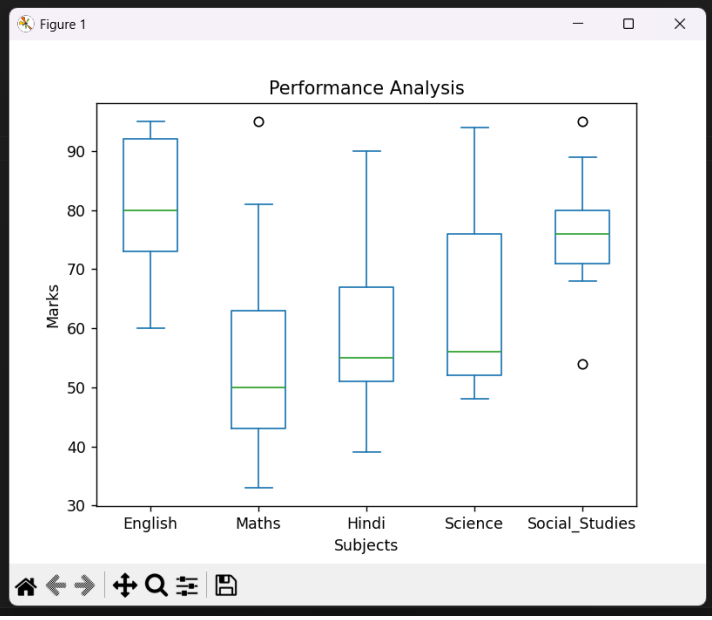
Q7.

Name	English	Maths	Hindi	Science	Social_Studies
Rishika Batra	95	95	90	94	95
Waseem Ali	95	76	79	77	89
Kulpreet Singh	78	81	75	76	88
Annie Mathews	88	63	67	77	80

In order to assess the performance of students of a class in the annual examination, the class teacher stored marks of the students in all the 5 subjects in a CSV "Marks.csv" file as shown in Table. Plot the data using boxplot and perform a comparative analysis of performance in each subject.

Ans.

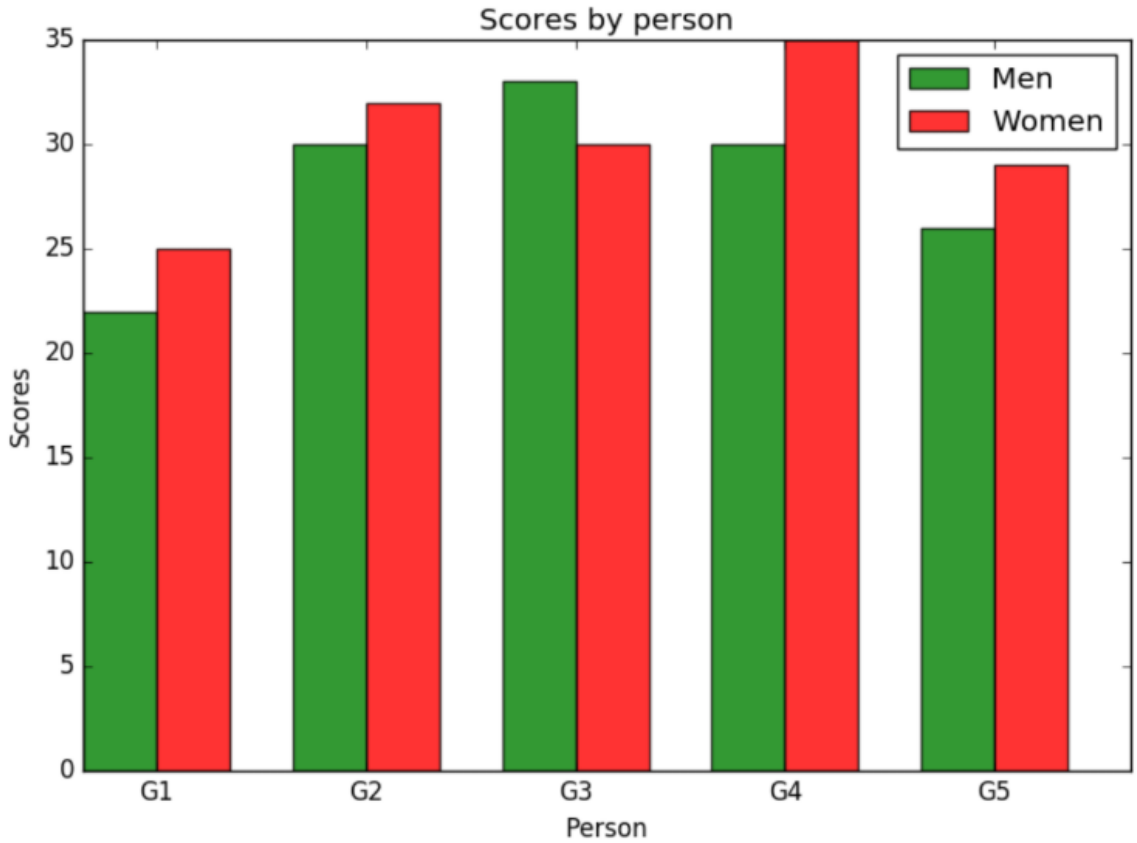
```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
data= pd.read_csv('Marks.csv')
df= pd.DataFrame(data)
df.plot(kind='box')
#set title,xlabel,ylabel
plt.title('Performance Analysis')
plt.xlabel('Subjects')
plt.ylabel('Marks')
plt.show()
```



Q8.

Write a Python program to create bar plot of scores by group and gender.
Use multiple X values on the same chart for men and women.

Sample Data:
Means (men) = (22, 30, 35, 35, 26)
Means (women) = (25, 32, 30, 35, 29)
The code snippet gives the output shown in the following screenshot:



Ans.

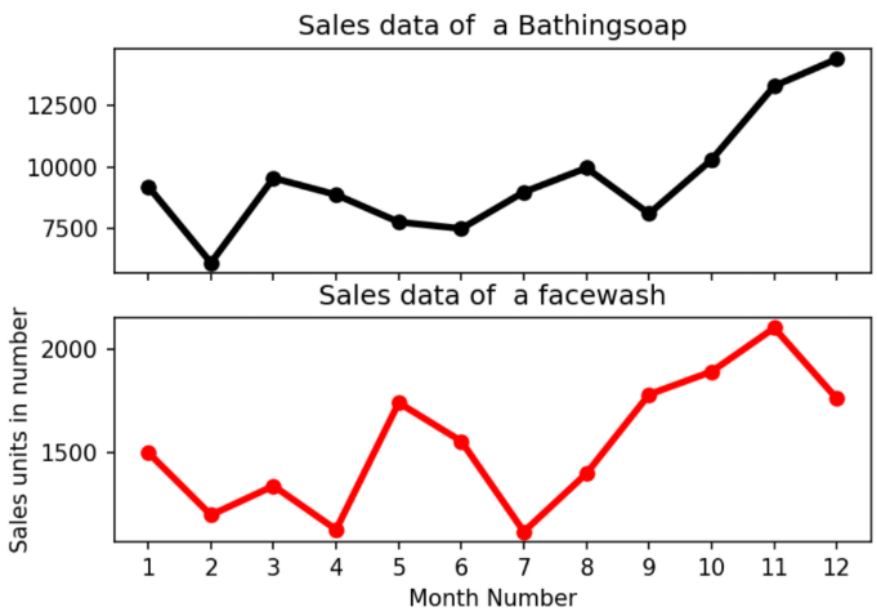
```
# create plot
fig, ax = plt.subplots()
index = np.arange(n_groups)
bar_width = 0.35
opacity = 0.8

rects1 = plt.bar(index, men_means, bar_width,
alpha=opacity,
color='g',
label='Men')

rects2 = plt.bar(index + bar_width, women_means, bar_width,
alpha=opacity,
color='r',
label='Women')

plt.xlabel('Person')
plt.ylabel('Scores')
plt.title('Scores by person')
plt.xticks(index + bar_width, ('G1', 'G2', 'G3', 'G4', 'G5'))
plt.legend()

plt.tight_layout()
plt.show()
```

Q9.

Use the following CSV file for this exercise
https://drive.google.com/file/d/1FTV7KLn_PzOjVHrT9--5hMaPKDfmr13X/view?usp=sharing

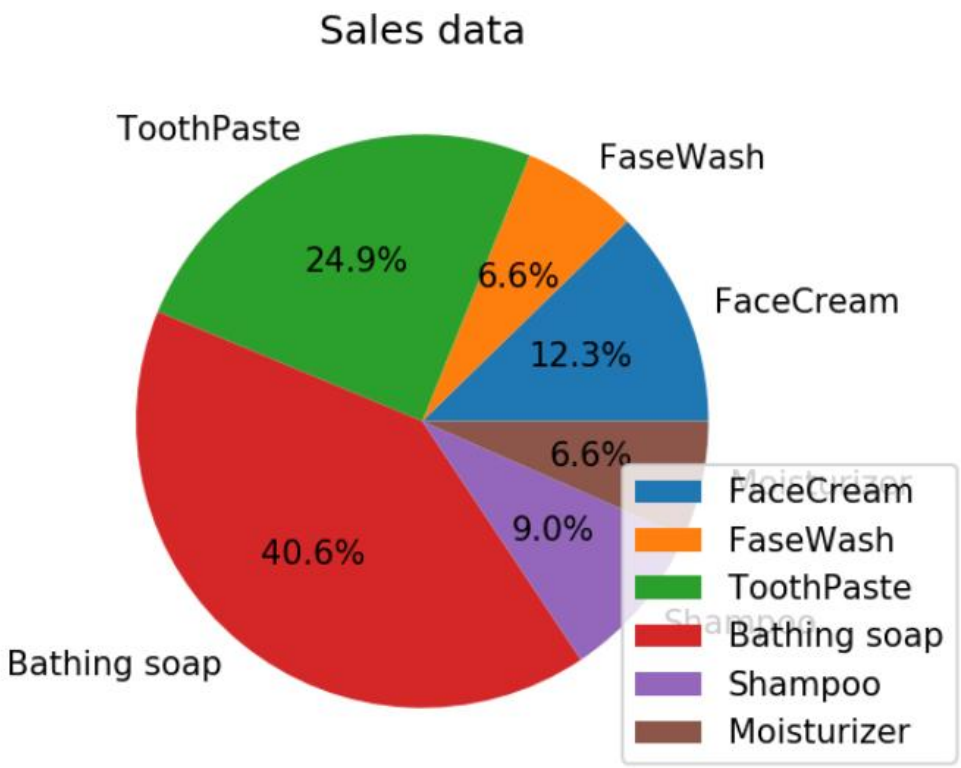
a. Calculate total sale data for last year for each product and show it using a Pie chart
   Note: In pie chart display Number of units sold per year for each product in percentage

b. Read Bathing soap facewash of all months and display it using the Subplot The Subplot should look like this.

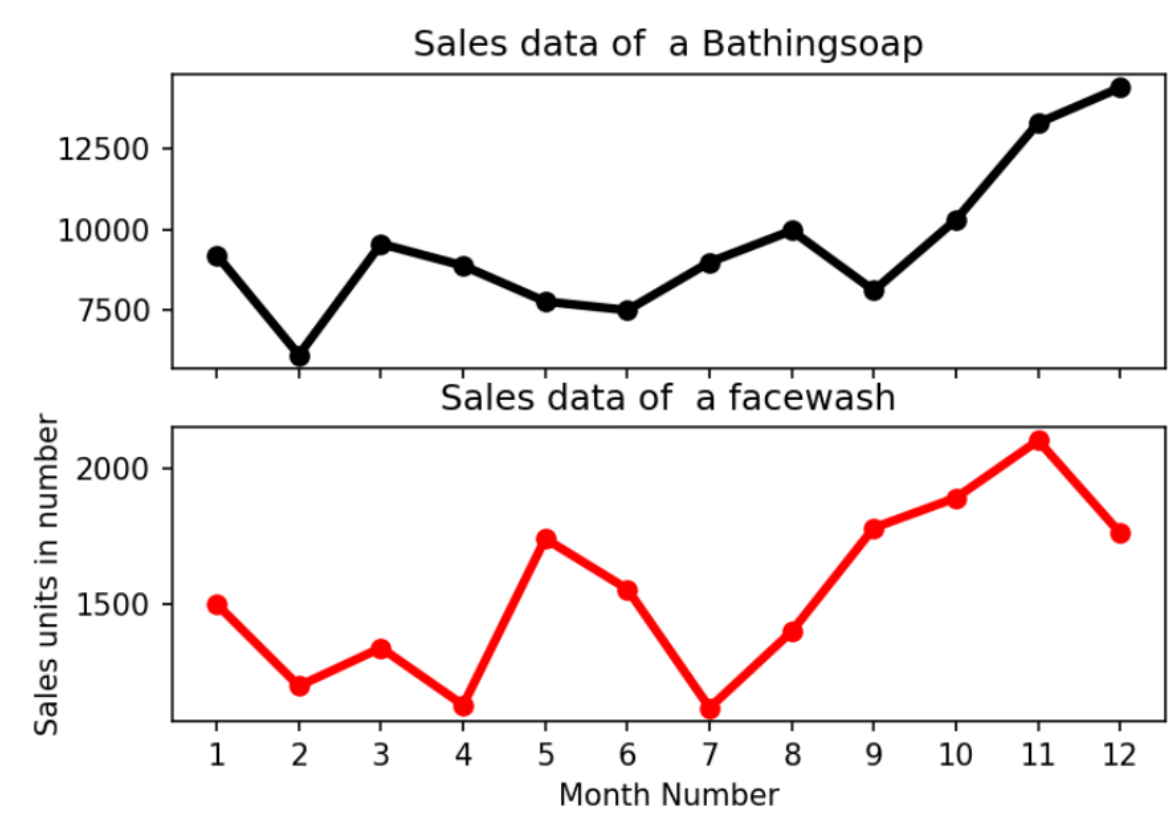


Ans.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("D:\\Python\\Articles\\matplotlib\\sales_data.csv")
monthList  = df ['month_number'].tolist()

labels = ['FaceCream', 'FaseWash', 'ToothPaste', 'Bathing soap', 'Shampoo', 'Moisturizer']
salesData   = [df ['facecream'].sum(), df ['facewash'].sum(), df ['toothpaste'].sum(),
         df ['bathingsoap'].sum(), df ['shampoo'].sum(), df ['moisturizer'].sum()]
plt.axis("equal")
plt.pie(salesData, labels=labels, autopct='%1.1f%%')
plt.legend(loc='lower right')
plt.title('Sales data')
plt.show()
```

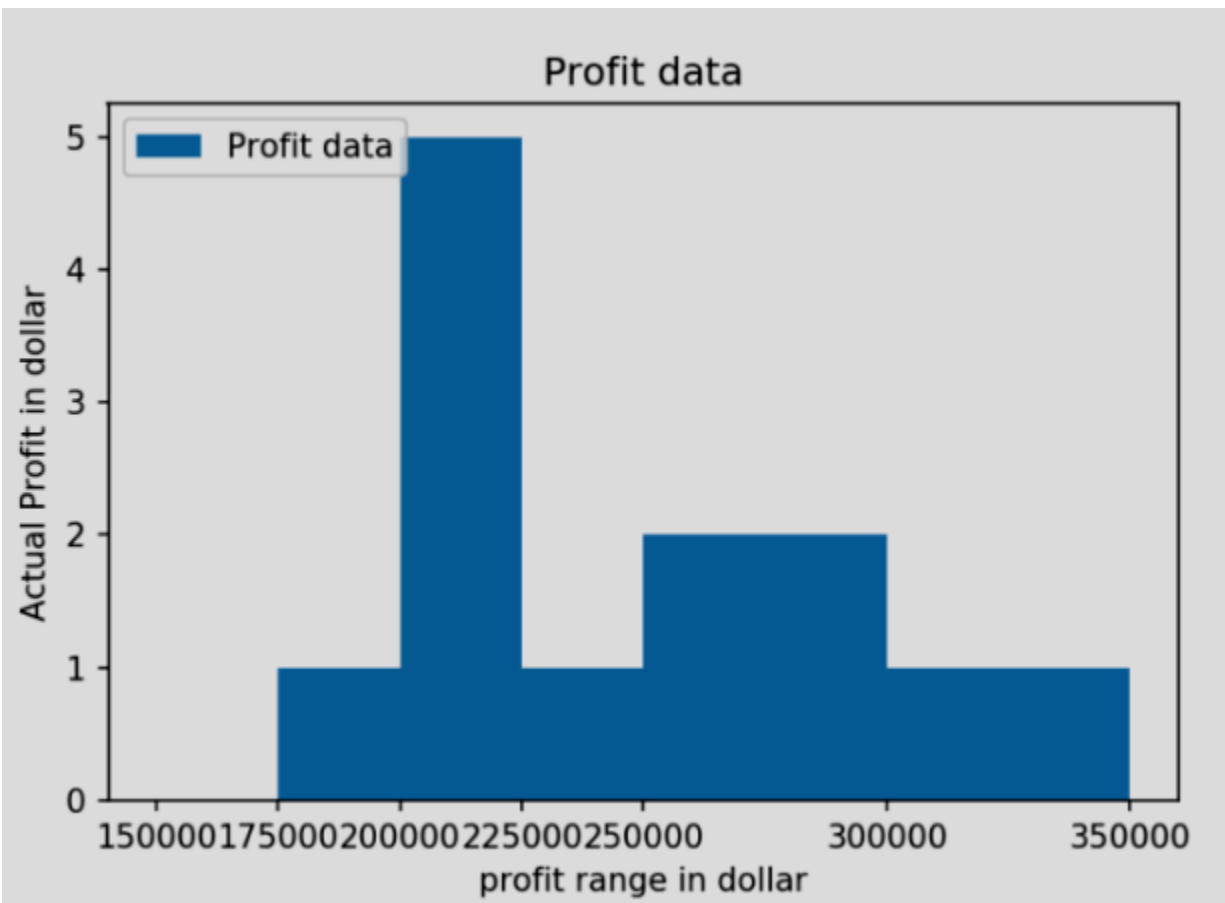


```python
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("D:\\Python\\Articles\\matplotlib\\sales_data.csv")
monthList  = df ['month_number'].tolist()
bathingsoap   = df ['bathingsoap'].tolist()
faceWashSalesData   = df ['facewash'].tolist()

f, axarr = plt.subplots(2, sharex=True)
axarr[0].plot(monthList, bathingsoap, label = 'Bathingsoap Sales Data', color='k', marker='o', linewidth=3)
axarr[0].set_title('Sales data of  a Bathingsoap')
axarr[1].plot(monthList, faceWashSalesData, label = 'Face Wash Sales Data', color='r', marker='o', linewidth=3)
axarr[1].set_title('Sales data of  a facewash')

plt.xticks(monthList)
plt.xlabel('Month Number')
plt.ylabel('Sales units in number')
plt.show()
```



Q10.

10.Read the total profit of each month and show it using the histogram to see the most common profit ranges

Ans.

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('company_sales_data.csv')
profitList = df ['total_profit'].tolist()
labels = ['low', 'average', 'Good', 'Best']
profit_range = [150000, 175000, 200000, 225000, 250000, 300000, 350000]
plt.hist(profitList, profit_range, label = 'Profit data')
plt.xlabel('profit range in dollar')
plt.ylabel('Actual Profit in dollar')
plt.legend(loc='upper left')
plt.xticks(profit_range)
plt.title('Profit data')
plt.show()
```